

Recursion (Assignment Solutions)

Question 1 :

```
int binSearch(int arr[], int si, int ei, int key) {  
    if(si > ei) {  
        return -1;  
    }  
  
    int mid = si + (ei - si)/2;  
    if(arr[mid] == key) {  
        return mid;  
    } else if(arr[mid] > key) { //left half call  
        return binSearch(arr, si, mid-1, key);  
    } else { //right half call  
        return binSearch(arr, mid+1, ei, key);  
    }  
}
```

Question 2 :

```
void allOccurrences(int arr[], int key, int i, int n) {  
    if(i == n) {  
        return;  
    }  
  
    if(arr[i] == key) {  
        cout << i << " ";  
    }  
  
    allOccurrences(arr, key, i+1, n);  
}
```

Question 3 :

```
int countSubstrs(string str, int i, int j, int n) {  
    if (n == 1) {
```

```
        return 1;
    }
    if (n <= 0) {
        return 0;
    }

    int res = countSubstrs(str, i + 1, j, n - 1) +
              countSubstrs(str, i, j - 1, n - 1) -
              countSubstrs(str, i + 1, j - 1, n - 2);

    if (str[i] == str[j]) {
        res++;
    }
    return res;
}

int main() {
    string str = "abcbab";
    int n = str.size();
    cout << countSubstrs(str, 0, n-1, n) << endl;
    return 0;
}
```

Question 4 :

The Solution for this particular question has also been discussed here in Java :
<https://www.youtube.com/watch?v=u-HgzgYe8KA>

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```
void towerOfHanoi(int n, string src, string helper, string dest) {
    if(n == 1) {
        cout << "transfer disk " << n << " from " << src << " to " << dest << endl;
        return;
    }
    //transfer top n-1 from src to helper using dest as 'helper'
    towerOfHanoi(n-1, src, dest, helper);
    //transfer nth from src to dest
    cout << "transfer disk " << n << " from " << src << " to " << helper << endl;
    //transfer n-1 from helper to dest using src as 'helper'
```

```
    towerOfHanoi(n-1, helper, src, dest);
}

int main() {
    int n = 4;
    towerOfHanoi(4, "A", "B", "C");
    return 0;
}
```

Question 5 :

```
long long power(long long a, long long b) {
    if(b==0) return 1;

    long long half_power= power(a,b/2);
    if(b%2 ==0)
        return half_power*half_power % MOD;

    else
        return half_power*half_power % MOD * (a % MOD) % MOD ;
}

int countGoodNumbers(long long n) {
    long long ed;
    long long od;
    if(n&1){
        od=n/2;
        ed=(n+1)/2;
    }
    else{
        od=n/2;
        ed=n/2;
    }
    return ( (power(5,ed)%MOD) * (power(4,od)%MOD) ) %MOD;
}
```